Systemic immunity-enhancing effects in healthy subjects following dietary consumption of the lactic acid bacterium Lactobacillus rhamnosus HN001.

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OBJECTIVE: To determine the effects of the probiotic lactic acid bacterium, Lactobacillus rhamnosus HN001, on natural cellular immunity when delivered orally in normal low-fat milk (LFM) or lactose-hydrolyzed low-fat milk (LFM-LH). DESIGN: A three stage, pre-post intervention trial, spanning nine weeks. SETTING: Taipei Medical College Hospital, Taipei, Taiwan. SUBJECTS: Fifty-two healthy middle-aged and elderly volunteers (17 males, 35 females; median age 63.5, range 44-80). INTERVENTIONS: Stage 1 (run-in diet): 25 g/200 mL reconstituted LFM powder, twice daily for 3 weeks. Stage 2 (probiotic intervention): LFM or LFM-LH, supplemented with 10(9) CFUs/g L. rhamnosus HN001 in each case, for 3 weeks. Stage 3 (wash-out): LFM for 3 weeks. MEASURES OF OUTCOME: In vitro phagocytic capacity of peripheral blood polymorphonuclear (PMN) leukocytes; in vitro tumoricidal activity of natural killer (NK) leukocytes. RESULTS: Immunological responses were unaffected by the run-in diet of LFM alone. In contrast, the relative proportion of PMN cells showing phagocytic activity increased by 19% and 15%, respectively, following consumption of HN001 in either LFM or LFM-LH; the relative level of NK cell tumor killing activity increased by 71% and 147%. In most cases these levels declined following cessation, but remained above baseline. CONCLUSIONS: Dietary consumption of L. rhamnosus HN001, in a base of low-fat milk or lactose-hydrolyzed low-fat milk, appears to enhance systemic cellular immune responses and may be useful as a dietary supplement to boost natural immunity.